4 x 4 analog matrix
Schematic revision & SPICE checks

REV A2 | 30 SEPTEMBER 2026 | LTspice 24.1.3 / KiCad 8.0.6

Result: the revised nominal model supports DC-to-1 MHz amplitude operation into 50 ohms. Simulation found a weak input-buffer stability margin, which was improved by changing the 32 polarity-branch resistors from 49.9 ohms to 150 ohms. It does **not** establish 1% relative accuracy for every coefficient or hardware stability under every load.

Final A2 check	Result	Interpretation
1 MHz amplitude, DC-normalized	-0.758%	Nominal full matrix; all outputs loaded
Nominal -3 dB frequency	7.24 MHz	First sampled crossing; approximate
Loaded polarity-buffer margins	61.2 to 65.5 degrees	Feedback-input injection; typical models
Nominal DAC/TIA margin	68.7 degrees	40 pF output node, 2.5 pF total feedback
Full-scale 1 MHz sine	3.971 V peak	Four matched-input row surrogate
0-to-4 V output step	119 ns to 1%	50 ohms || 20 pF; no visible overshoot
1 nF load step	Unresolved	Numerical stall; not a passing test
1% relative coefficient accuracy	Not established	Illustrative feedthrough defeats small M

Why the earlier schematic looked scattered

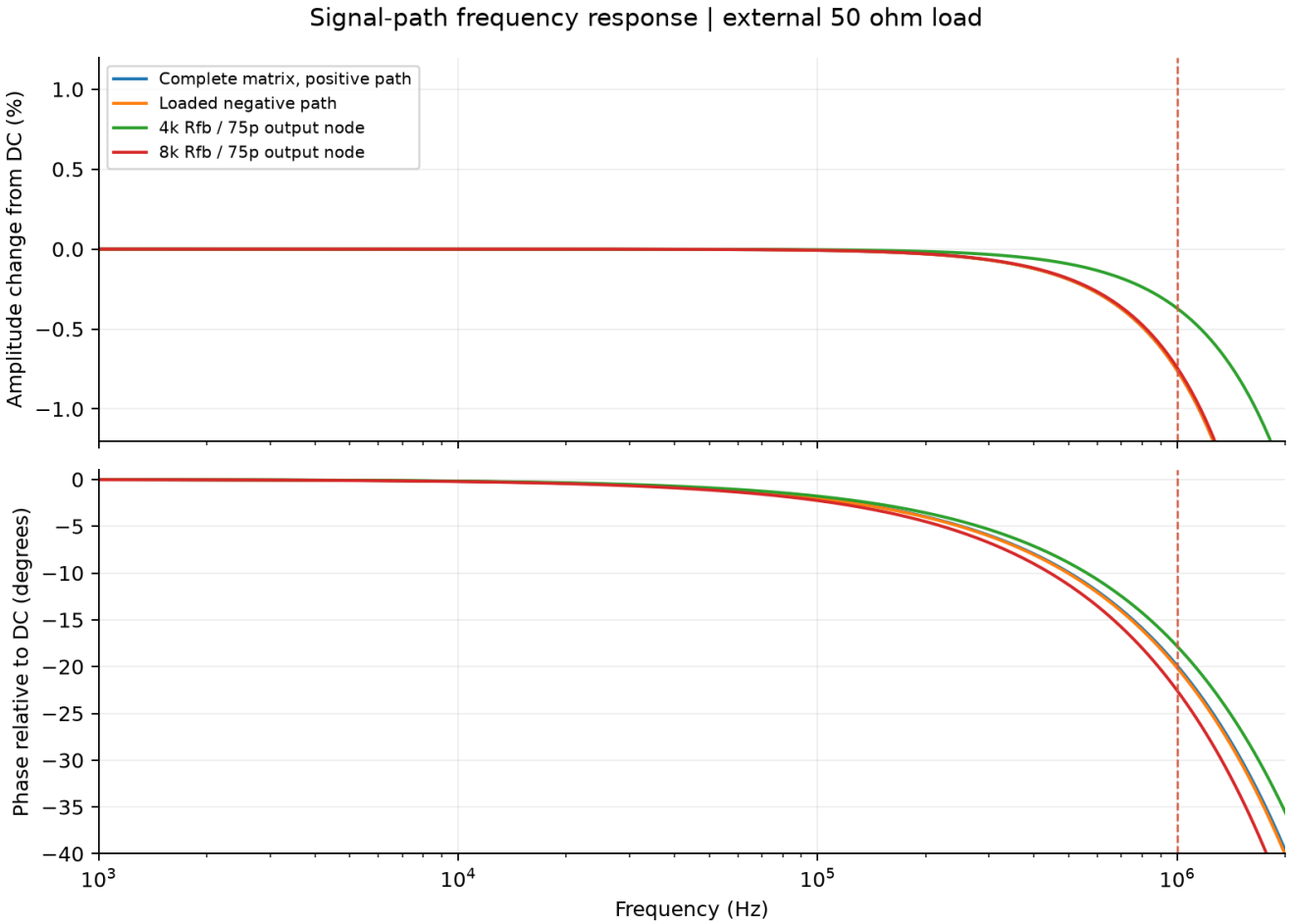
The first conversion split input channels, weights, summing stages and power units over 16 sheets, forcing too much tracing through labels. A2 uses **nine sheets**: an index, all four input lanes, four complete output rows, offset trim, control and power. Each row now has continuous wires from its four weight cells through the summer to its output connector; local power units sit beside their circuits.

The AD5544BRSZ package was also corrected to **SSOP-28, 5.3 x 10.2 mm body, 0.65 mm pitch**. The earlier TSSOP footprint was wrong even though its pad numbers matched. The GPIO mapping is unchanged; the firmware default gain estimate is updated to 1.021 and still requires measured calibration.

Final KiCad checks: 428 components; 18 local footprint types; zero ERC violations, zero pin/net mismatches and zero symbol-pin/footprint-pad mismatches. This package contains an editable schematic, not a routed PCB.

1. Frequency response and calibrated gain

The complete analog matrix includes all 16 weight cells and four terminated outputs. One input is driven; the other three are grounded. Coefficients are full-code positive, controls are static and trim outputs are held at ideal -1.25 V. The multiplying DACs use the approximate equivalent described on page 6.



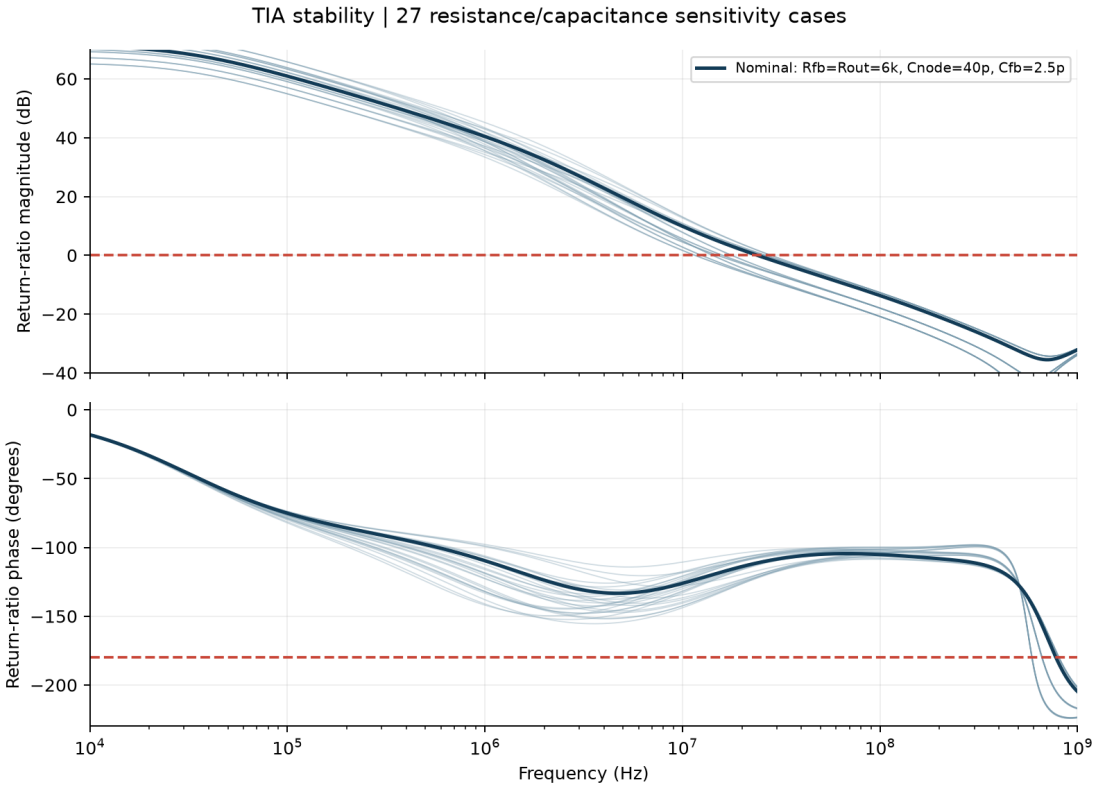
Quantity at 1 MHz	Positive path	Negative path
Amplitude relative to DC	0.992420	0.992407
Phase relative to DC	-19.89 degrees	-20.13 degrees
Uncalibrated low-frequency full-code magnitude	1.020861 V/V	1.020844 V/V

Gain normalization removes the fixed full-code gain error using a coefficient adjustment. It does not remove phase delay. The two polarity paths differ by about 0.23 degrees at 1 MHz in the matched model; channel/component mismatch and cable delay are additional effects. These results address **amplitude flatness**, not 1% instantaneous or complex-valued matrix accuracy.

The two resistance/capacitance curves are sensitivity cases, not guaranteed process or temperature corners. They vary DAC reference/feedback resistance, assumed output resistance and capacitance. A complete tolerance and thermal error budget remains necessary. AC sweeps start at 10 Hz as a practical DC limit; a separate DC sweep checks headroom.

2. Loop stability and the resistor correction

Loop / case	Estimated phase margin
Input unity buffer, final 150-ohm fanout	59.0 degrees
Positive polarity buffer: 49.9 ohms -> 150 ohms	33.0 -> 61.2 degrees, selected-path load
Negative polarity buffer, final sign-state extremes	62.7 to 65.5 degrees
Summer / offset trim	70.6 / 79.2 degrees
TIA nominal / lowest of 27 sensitivity cases	68.7 / 44.6 degrees



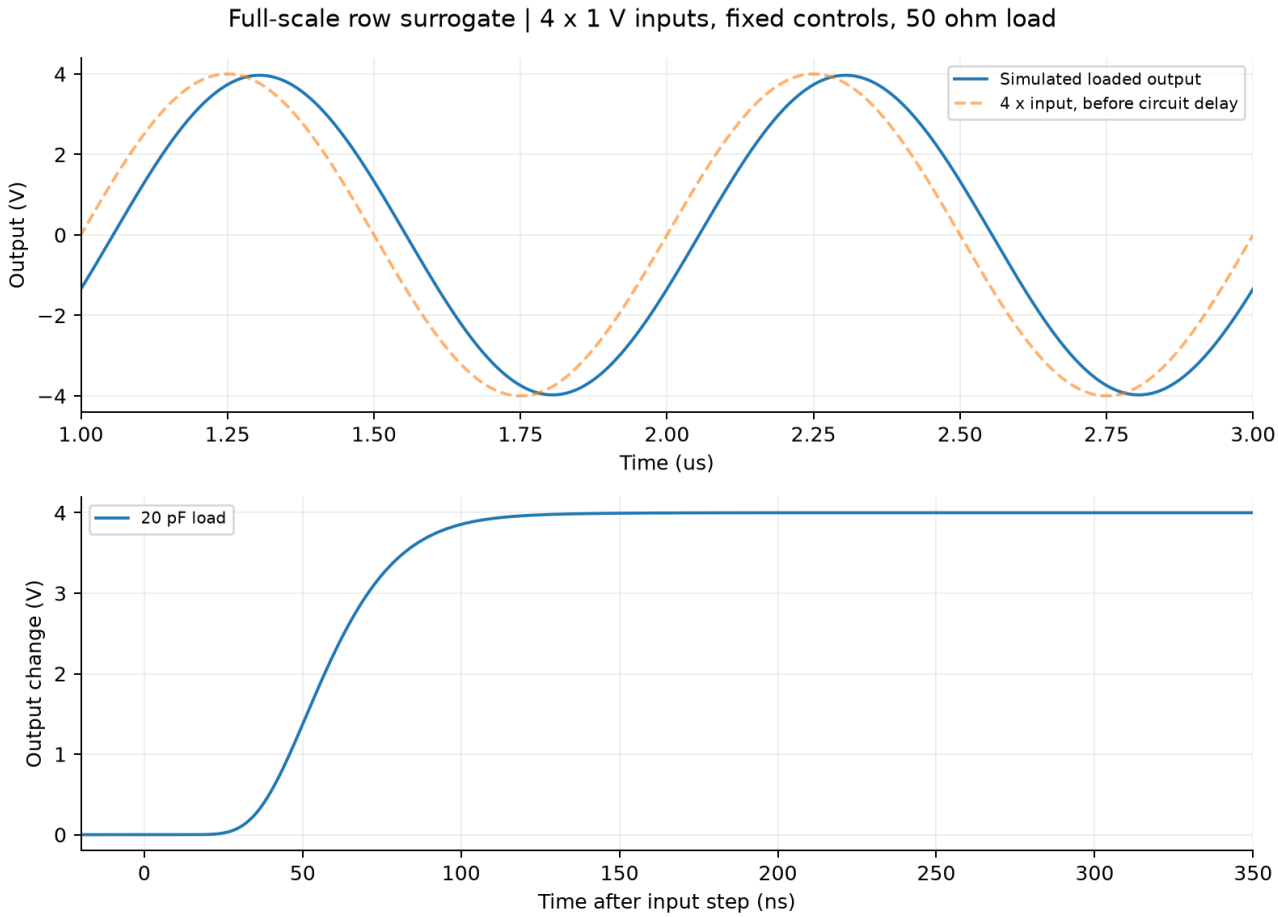
The 27 TIA cases combine RFB = 4k, 6k or 8k; assumed Rout = 2k, 6k or 1G; and output-node C = 35, 70 or 100 pF. Feedback is 2.2 pF plus 0.3 pF assumed stray. The 100 pF stress cases reduce margin to about 45 degrees. Keep the IOUT/feedback layout short and reserve the feedback capacitor for bench tuning.

These are single-injection return-ratio estimates, T = -V(raw)/V(out), with DC feedback closed. Input-stage injection is at the feedback input; TIA, summer and trim injection is at the output. An initial polarity-buffer output-injection setup produced extra GHz crossings and is not used for the reported buffer margins. Injection-point impedance separation limits the method.

The THS3091 model reports about 70.7 degrees in its load sweep, but its own header says open-loop gain is not accurate. That number is only a model diagnostic. Keep the 1.21k feedback resistor, avoid a feedback capacitor on this current-feedback amplifier, and verify the actual driver/load on hardware.

3. Output headroom and time-domain checks

One complete, loaded input/weight chain drives a summer together with three ideal copies of its weight waveform. This models four matched, in-phase inputs while preserving summer noise gain and the modeled TIA load. It is **not** a full four-independent-channel transient. Amplifier models are unchanged; switches use the static RC reduction described on page 6.



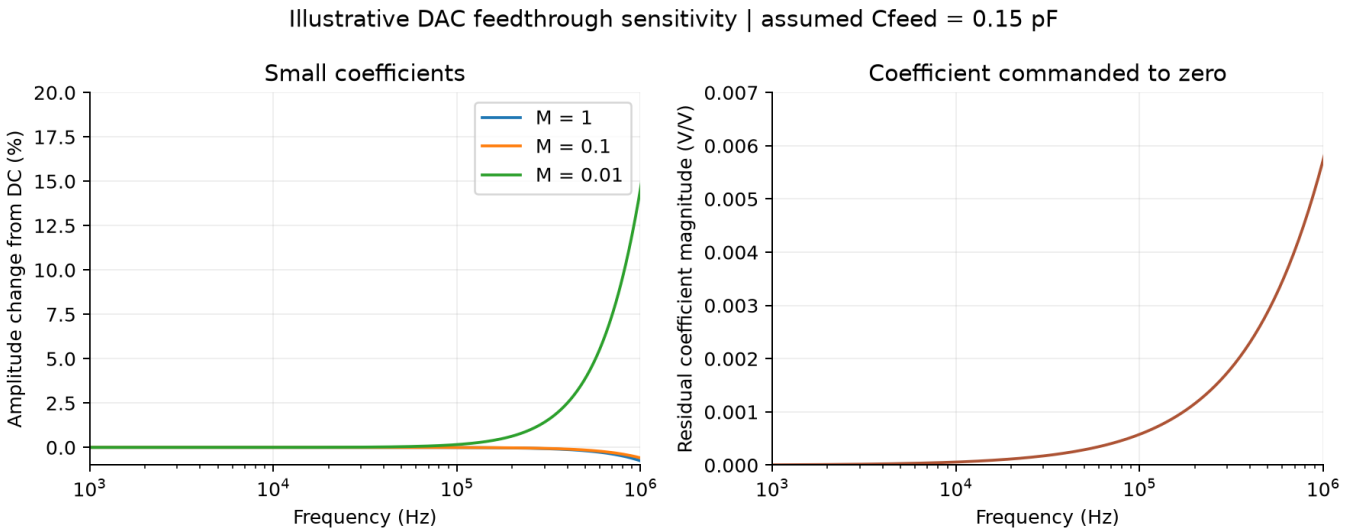
Measured quantity	Final A2 result
1 MHz sine, nominal sum of four 1 V inputs	3.9705 V peak; -19.72 degrees phase
Step, 10 ns input edge, 50 ohms || 20 pF	3.9999 V; 119.3 ns to 1%
DC sweep, each matched input from -1 to +1 V	-4.0036 to +3.9958 V
DC fitted offset, trim held at midpoint	-3.94 mV; calibratable
1 nF external load step	Unresolved: timestep stall / bounded-run timeout

A 4 V loaded output requires about 80 mA and approximately 8 V at the amplifier output because of the 49.9-ohm source resistor. Nominal simulated headroom is adequate on +/-15 V rails. The model does not certify THD, thermal behavior, startup sequencing, cable reflections or simultaneous independent-channel transients.

The step-settling time is measured from the start of the 10 ns input edge. The displayed trace covers the rising edge; the delivered complete trace also contains the falling edge. No incomplete 1 nF trace is used in the measurements.

4. What the 1% accuracy target means

An ideal scaled-current DAC model makes normalized bandwidth look almost independent of coefficient. That is optimistic at small settings. To expose the sensitivity, a 0.15 pF VREF-to-IOUT capacitor was added. With 6k feedback it approximates the AD5544 data sheet's typical zero-code feedthrough of -65 dB at 100 kHz. It is **an illustrative assumption, not a measured capacitance or worst-case bound**.



DC-calibrated target M	Magnitude at 1 MHz	Amplitude error
1.0	0.992440	-0.756% relative
0.1	0.099408	-0.592% relative
0.01	0.011459	+14.59% relative
0	0.005729 V/V	Residual input coupling; relative error undefined

If 1% means 1% of each nonzero coefficient, this sensitivity example fails badly for M = 0.01. Static MCU gain calibration cannot remove frequency-dependent feedthrough. Deep cancellation also needs phase matching, not just accurate amplitudes.

If 1% means **an absolute coefficient error of +/-0.01**, the zero-code residual in this one nominal example is below 0.01. That alone is not a pass for every coefficient: amplitude, phase, tolerances, feedthrough and contributions from four inputs must be combined. Relative error in a nearly cancelled output can still be very large.

For tighter small-coefficient or cancellation requirements, characterize a single populated weight cell over code and frequency before committing the full PCB. This revision deliberately retains the uncertainty instead of presenting the ideal-DAC curve as a guarantee.

Source: Analog Devices AD5544/AD5554 data sheet, dynamic characteristics and multiplying-mode test conditions.

5. Model scope, reproducibility and remaining checks

Part / function	Model used
OPA810 / OPA197	Official TI macromodels, SBOMB73 / SBOMA34
ADA4899-1	Unmodified encrypted model from installed ADI LTspice library
THS3091	Official TI TINA model, SLOM418; see accuracy limits below
TMUX6219, AC and DC	Original official TI model, SCDM270
TMUX6219, static waveforms	Authored Ron(V) and terminal-capacitance reduction
AD5544	Approximate controlled-current source plus R/C ladder equivalent
Rails, reference, control and full-matrix trim	Ideal sources; trim amplifier checked separately

The DAC equivalent uses nominal RREF = RFB = 6k, assumed Rout = 6k, 35 pF typical output capacitance plus 5 pF layout stray, and 2.2 pF external feedback plus 0.3 pF stray. Amplifier input capacitance is already in its model. No separate 12 MHz DAC pole is added: the published bandwidth already includes its test amplifier and compensation. Code-dependent impedance, INL, glitches and temperature behavior are omitted.

The static switch reduction retains the original +/-5 V Ron(V) table and terminal loading, including the static capacitance of omitted charge-injection blocks. Its loaded-path AC result differs from the original by at most 0.027% in amplitude and 0.184 degrees through 1 MHz. It does not simulate sign/mute transitions, charge injection, ESD or supply current.

The original switch transient model and full matrix with nonlinear trim caused convergence problems. Successful full-matrix AC uses fixed trim sources; the independent trim DC result is -1.249961 V at midcode. Failed variants and the unresolved 1 nF stress case are identified in the manifest. The THS3091 model does not accurately model open-loop gain or distortion and has bandwidth/slew limitations, so physical output-stage validation remains essential.

Reproduce and use the files

Open the A2 KiCad project with its local libraries. Open any delivered spice/benches/*.cir in LTspice and run it, or run LTspice.exe -b filename.cir from the benches directory. Keep the relative models/ folder intact. Complete traces, log excerpts, model provenance, hashes and validation_results.json accompany the report. analyze.py requires NumPy; plot_results.py also requires Matplotlib. MODEL_SCOPE.txt contains the full assumptions.

Before routing: confirm the required coefficient-error definition and characterize small-code feedthrough. After layout: check the TIA feedback capacitance, output-driver step response with the intended cable/load, regulator/driver heat, and both-sign calibration over frequency. The MCU protocol and power-up sequence have not been verified by analog SPICE.

Primary model sources

OPA810: https://www.ti.com/lit/zip/SBOMB73

OPA197: https://www.ti.com/lit/zip/SBOMA34

THS3091: https://www.ti.com/lit/zip/SLOM418

TMUX6219: https://www.ti.com/lit/zip/SCDM270

ADA4899-1: https://www.analog.com/en/products/ada4899-1.html